Which Collision Laws Admit Finite-Moment Prediction? A Gaussian-Mixture Classification and a Clock-Free Inverse

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Conditional classification with exact controls; no physical-gas validation.
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Abstract

For homogeneous isotropic elastic collisions with uniform outgoing relative direction, the initial fourth velocity-moment response of a centered Gaussian mixture is a quadratic functional of its temperature law. We characterize when that functional is exactly determined by any finite list of temperature moments, allowing an arbitrary determining function. For moments through order $m \geq 2$, this occurs precisely when the thermal collision response is a polynomial of degree at most $m-2$. Under stated integrability assumptions, this is equivalent to a radial collision kernel polynomial in squared relative speed. The statement concerns one initial response, not autonomous closure of a full moment hierarchy. On a known compact temperature interval, polynomial approximation also bounds initial-response prediction error and the ambiguity between preparations sharing the retained moments. In a separately postulated Maxwell-type moment model, a two-time logarithmic ratio cancels an unknown changing collision clock. It identifies a constant angular coordinate, but only its exposure-weighted mean when the law varies. An exact counterexample shows why finitely many endpoints cannot determine the intervening law. No laboratory validation is involved.

1 Scope and established baseline

Closed Maxwell-molecule moment hierarchies are classical, with exact solutions and modern finite-factor formulations [4, 5]. The historical comparators are the Maxwell-molecule moment calculations of Ikenberry and Truesdell, Grad's moment method, and the exact Boltzmann solution of Krook and Wu [1, 2, 3]. They are baselines, not consequences of the present classification. Polynomial moment preservation also has a broad independent literature [6]. Neither the existence of Maxwell closure nor the inadequacy of an arbitrary truncation is offered as a new result here. Entropy-based moment-closure hierarchies [9] concern a different task from exact constancy of one response over all measures with prescribed moments; this theorem does not rule out those closures as useful approximations. The truncated moment problem asks which measures represent prescribed moments [7]; here the question is constancy of a particular quadratic functional across all such measures. Laplace-transform uniqueness is also established machinery [8], not a claimed new inverse theorem.

Our first question is more restricted and has a converse: which collision responses can be predicted exactly from finitely many instantaneous moments throughout the entire class of positive-temperature Gaussian mixtures? Our second question concerns a different observation resource: what does a two-time ratio identify if the collision rate or angular law changes? Combining these questions makes explicit the distinction between a state summary, a dynamical law and an observation history.

2 Preparations and the initial response

Let μ be a compactly supported probability measure on $(0,\infty)$ and

$$f_\mu(v)=\int (2\pi\theta)^{-3/2}\exp(-|v|^2/(2\theta))\,\mu(\,\mathrm{d}\theta),\qquad M_j=\int \theta^j\,\mu(\,\mathrm{d}\theta).$$

Then $\int |v|^{2j}f_\mu(v)\,\mathrm{d}v=(2j+1)!!M_j$. Temperature moments through order m correspond to radial velocity moments through degree $2m$.

Fix the collision normalization explicitly. For a test function ψ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\psi\rangle=\frac{1}{2}\iint f(v)f(w)K(|v-w|)\int_{S^2}[\psi(v')+\psi(w')-\psi(v)-\psi(w)]\frac{\mathrm{d}\sigma}{4\pi}\,\mathrm{d}v\,\mathrm{d}w,$$

where $V=(v+w)/2$, $g=v-w$ and $v',w'=V\pm|g|\sigma/2$. Thus D below is the derivative of the raw moment $\langle|v|^4\rangle$, not of that moment divided by 15. For this model the initial fourth velocity-moment derivative is

$$D(\mu)=-\iint(\theta-\eta)^2\Phi(\theta+\eta)\,\mu(\,\mathrm{d}\theta)\mu(\,\mathrm{d}\eta). \tag{1}$$

For a radial speed factor K, the thermal response is

$$\Phi(s)=\frac{s^{-7/2}}{6\sqrt{2\pi}}\int_0^\infty x^{5/2}K(\sqrt{x})\mathrm{e}^{-x/(2s)}\,\mathrm{d}x. \tag{2}$$

These formulas assume elastic collisions and a uniform outgoing relative direction. A nonuniform angle law is not silently absorbed into a claim about the bare radial kernel. For completeness, conditional on temperatures θ,η, write $s=\theta+\eta$. Then g is Gaussian with covariance sI, and $V\mid g=(\theta-\eta)g/(2s)+\zeta$, with independent centered ζ of covariance $\theta\eta I/s$. The angularly averaged weak increment for the fourth moment is $K(|g|)\{|g|^2|V|^2/3-(V\cdot g)^2\}$. Conditioning on g gives $-(\theta-\eta)^2K(|g|)|g|^4/(6s^2)$. The Maxwell radial density and the substitution $x=|g|^2$ give (2), including its normalization.

Definition 2.1. The response is *determined through order m* if any two admitted probability measures sharing $M_0,\dots,M_m$ have the same D. No continuity, polynomial form or linearity of the determining map is assumed.

3 An exact finite-moment classification

Theorem 3.1. *Let $\Phi\in C^\infty((0,\infty))$ and $m\ge 2$. On all compactly supported probability measures on $(0,\infty)$, the following are equivalent:*

(i) D is determined through order m.

(ii) Φ is a polynomial of degree at most $m-2$.

For a nonzero polynomial of degree J, the least cutoff is $J+2$. For a nonpolynomial smooth Φ, no finite cutoff works on this class.

Proof. Write $h(x,t)=(x-t)^2\Phi(x+t)$ and $B(\rho,\nu)=\iint h(x,t)\rho(\,\mathrm{d}x)\nu(\,\mathrm{d}t)$.

Suppose (i) holds. Let ν be any finite signed atomic measure with vanishing moments through order m. Choose a probability measure ρ giving positive mass to every support point of ν. For small

$\varepsilon > 0$, both $\rho + \varepsilon\nu$ and $\rho - \varepsilon\nu$ are probability measures with the same moments. The difference of their quadratic functionals is $-4\varepsilon B(\rho, \nu)$, so $B(\rho, \nu) = 0$.

For any $t > 0$ and $0 < a < 1$, the measure $\rho_a = (1 - a)\rho + a\delta_t$ still permits both signed perturbations. Thus $B(\rho_a, \nu) = 0$ and consequently $B(\delta_t, \nu) = 0$. By symmetry of h, this says that $h(\,\cdot\,, t)$ annihilates every such signed measure. In particular, the $(m + 1)$st forward difference in x vanishes, since its binomial signed measure annihilates all monomials through order m. Taking its mesh to zero yields $\partial_x^{m+1} h(x, t) = 0$ for every $x, t > 0$. At $x = t$ the Leibniz rule leaves only

$$0 = m(m + 1)\Phi^{(m-1)}(2t).$$

Connectedness of the positive half-line proves (ii). This argument never differentiates the unknown determining map.

Conversely, for $\Phi(s) = \sum_{j=0}^{J} c_j s^j$ direct expansion gives

$$D = -\sum_{j=0}^{J} c_j \sum_{\ell=0}^{j} \binom{j}{\ell} \left[M_{\ell+2} M_{j-\ell} - 2M_{\ell+1} M_{j-\ell+1} + M_\ell M_{j-\ell+2} \right]. \tag{3}$$

Only moments through $J + 2$ occur. Necessity excludes any smaller cutoff of at least two. A cutoff of one cannot suffice for any nonzero Φ: δ_a and $(\delta_{a-h} + \delta_{a+h})/2$ have the same mean but responses 0 and $-2h^2\Phi(2a)$, respectively. Choose a with $\Phi(2a) \neq 0$ and $0 < h < a$. □

The supplement [10] checks the moment formula and separating examples exactly; these finite checks do not replace the proof of the universally quantified statement.

Corollary 3.2 (Approximate initial-response prediction)**.** *Let $0 < a < b$, $m \geq 2$, $\Phi \in C([2a, 2b])$, and let μ be a probability measure on $[a, b]$ with temperature variance $v = M_2 - M_1^2$. For a polynomial P of degree at most $m - 2$, put*

$$\varepsilon(P) = \|\Phi - P\|_{\infty,[2a,2b]}, \qquad D^P(\mu) = -\iint (\theta - \eta)^2 P(\theta + \eta)\, \mu(\,\mathrm{d}\theta)\mu(\,\mathrm{d}\eta).$$

The polynomial predictor $D^P(\mu)$ is determined by $M_0, \ldots, M_m$ through (3), and

$$|D(\mu) - D^P(\mu)| \leq 2v\varepsilon(P).$$

If ν is another probability measure on the same interval with those moments, then, writing $\varepsilon_m = \inf_{\deg P \leq m-2} \varepsilon(P)$,

$$|D(\mu) - D(\nu)| \leq 4v\varepsilon_m \leq (b - a)^2 \varepsilon_m.$$

Proof. The polynomial moment formula gives $D^P(\mu) = D^P(\nu)$, while

$$\iint (\theta - \eta)^2\, \mu(\,\mathrm{d}\theta)\mu(\,\mathrm{d}\eta) = 2v.$$

Bounding the integrand error gives the single-predictor inequality. The triangle inequality and the common variance give the two-preparation bound after taking the infimum over P. Finally $\int (\theta - a)(b - \theta)\, \mu(\,\mathrm{d}\theta) \geq 0$ implies $v \leq (M_1 - a)(b - M_1) \leq (b - a)^2/4$. □

Thus $2v\varepsilon(P)$ controls a specified predictor, whereas $4v\varepsilon_m$ bounds the diameter of responses compatible with the retained moments. This elementary approximation consequence does not claim that the diameter bound is sharp. A numerical guarantee requires an actual polynomial and a certified uniform error bound on the stated support interval, not only the definition of ε_m.

3.1 What this theorem does not close

Even when (3) holds, the higher moments appearing in its right-hand side need not themselves have autonomous equations at that cutoff. The theorem addresses the initial derivative on a preparation class, not the invariance of that class under every subsequent collision evolution. It does not exclude approximate closure, restricted-family closure, extra measured observables or history-dependent inference. Corollary 3.2 concerns only the initial response; uncertain support, noisy moments and propagation to later times require separate error estimates.

4 Classification in the raw speed variable

Corollary 4.1. *Assume $K \geq 0$ is measurable and $\int_0^\infty r^6 K(r) \mathrm{e}^{-\lambda r^2}\,\mathrm{d}r < \infty$ for every $\lambda > 0$. Then the thermal response in (2) is a polynomial $\sum_{j=0}^{J} c_j s^j$ if and only if, almost everywhere on $r > 0$,*

$$K(r) = \sum_{j=0}^{J} \frac{6c_j}{(2j+5)!!} r^{2j}.$$

The resulting polynomial must be nonnegative on the positive half-line; its individual coefficients need not be nonnegative.

Proof. The Gamma integral maps $K(r) = r^{2j}$ to $\Phi(s) = (2j+5)!!\, s^j/6$. Subtract this finite polynomial from K when Φ is polynomial. The Laplace transform of $x^{5/2}$ times the difference at $\sqrt{x}$ vanishes for every positive transform parameter. The stated exponential integrability allows Laplace uniqueness, so the difference vanishes almost everywhere. □

The value at $r = 0$ is not identified. This is not a classification of anisotropic scattering or of all intermolecular force laws.

5 A clock-canceling inverse in a Maxwell model

Consider a separate admitted angular family with normalized moments $q_j = \langle |v|^{2j} \rangle / [(2j+1)!!\, T^j]$, where $T = \langle |v|^2 \rangle / 3$ is conserved, and equations

$$\begin{aligned} q_2' &= -\nu(t)(q_2 - 1)/2, \\ q_3' &= -3\nu(t)(q_3 - q_2)/4, \\ q_4' &= \nu(t)\left[-(6+G)q_4/8 + Gq_3/2 + 3(2-G)q_2^2/8\right]. \end{aligned} \tag{4}$$

Here $\nu \geq 0$ and νG are integrable on each observation interval, and $G(t)$ is an admitted angular coordinate. These equations, rather than an unqualified identification of G with a molecular parameter, define the inverse problem.

Set $y = q_2 - 1$ and $R = q_4 - 4q_3 + 3q_2^2$. Substitution gives

$$y' = -\nu y/2, \qquad R' = -\nu(6+G)R/8.$$

Theorem 5.1. *If $y, R > 0$ and $A = \int_{t_1}^{t_2} \nu(t)\,\mathrm{d}t > 0$, then*

$$\widehat{G} := 4\frac{\log(R(t_2)/R(t_1))}{\log(y(t_2)/y(t_1))} - 6 = \frac{\int_{t_1}^{t_2} \nu(t) G(t)\,\mathrm{d}t}{A}. \tag{5}$$

A constant G is recovered despite an unknown changing common clock. For varying G, only the collision-exposure-weighted mean is identified.

Proof. Integrate the two scalar logarithmic derivatives and eliminate A. □

The denominator and sign conditions are admission requirements. For zero exposure or a vanishing invariant the displayed inverse is undefined. If uncertainty intervals permit either case or an invalid logarithm, a central estimate alone cannot justify applying the inverse.

5.1 Exact endpoint ambiguity

Use the same initial q_2, q_3, q_4 and total exposure in two experiments. In the first take $G = 7/5$ throughout. In the second take $G = 6/5$ for the first half of exposure and $G = 8/5$ for the second. Both histories have the same exposure-weighted mean. The first two moment equations are independent of G; the equation for R depends only on its integrated value. Therefore every displayed endpoint moment is identical. An intermediate observation can distinguish these particular histories, but the same construction can be placed inside each interval of any finite schedule. This defeats unrestricted history identification, not constant-law inference.

Exact continuous data instead supply the exposure coordinate $s = -2\log(y/y(t_1))$, along which

$$\frac{\mathrm{d}\log R}{\mathrm{d}s} = -(6 + G(s))/8.$$

Recovering this derivative from noisy finite data is a separate stability problem, not solved by (5).

6 Predictive interpretation and a finite next test

A useful future experiment would first prepare two independently verified velocity distributions with the same admitted low moments. A specified ordinary collision law would predict their unequal initial fourth-moment responses before either slope was observed. A separate rate calibration, or the admitted two-time inverse, would constrain the changing clock.

The classification says when equality of finitely many summaries can guarantee equality of the initial response. It does not say how large a detectable separation is for every preparation or noise budget.

7 Reproducibility and remaining review

The supplement supplies an exact-rational counterexample with matching M_0, M_1, M_2, polynomial-response controls, the transform constants, and a symbolic check of the clock-canceling invariant and endpoint ambiguity. No private data or machine-specific path is required. Its finite checks complement the proofs; they do not establish the converse by enumeration. The cited works establish the relevant baselines; a systematic novelty comparison for this particular classification has not been made, and no priority is claimed.

Drafting and verification disclosure

Luqman Mahmood directed the research and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted mathematical work, computation, checking and editing. Statements have different evidence classes: written conditional proofs, computer-assisted certificates, numerical demonstrations, and prospective protocols. These are labelled as such in the text and do not constitute external peer review. No claim of literature priority follows from computational

verification. The paper and mathematical data are licensed CC BY 4.0. Software source and executable verification scripts are excluded from that grant, with rights reserved unless a file explicitly states otherwise.

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